Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/31/2005
PI Name:	Santhanakrishnan, Sivakumar Ph.D.		
Project Title:	Oxygen and Fuel Diffusion Flame Studies in Microgravity Motivated by Spacecraft Oxygen Storage Fire Safety		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sskris@iupui.edu	**Fax:**	FY 317 274-9744
PI Organization Type:	UNIVERSITY	**Phone:**	317 274-9711
Organization Name:	Indiana University - Purdue University Indianapolis		
PI Address 1:	723 W. Michigan St.		
PI Address 2:	SL 260H		
PI Web Page:			
City:	Indianapolis	**State:**	IN
Zip Code:	46202	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gore, Jay P (Purdue University) Sunderland, Peter B. (University of Maryland, College Park)		
Grant/Contract No.:	NNC04AA08A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Normal and inverse diffusion flames will be considered burning hydrocarbon fuel in oxidizers of 21-100% O2 in N2. Diagnostics will include color video, radiometry, and multi-wavelength soot pyrometry in normal gravity and microgravity. Tests in normal gravity also will be performed and will include measurements of soot extinction and spectral emissions. The measurements will be used to validate coupled soot chemistry and radiation computations. 1. Examine the effects of enhanced oxygen conditions and inverse burning on microgravity laminar gas jet diffusion flames. 2. Consider the implications of these burning conditions on spacecraft fire safety. 3. Perform computational studies to supplement the experimental work. One goal central to NASA’s mission is to improve spacecraft fire safety for space shuttles and the space station. The nearly catastrophic Mir fire started when a canister leaked heated oxygen into the cabin. This work is expected to reveal new insights into possible fire scenarios on spacecraft and may lead to improved ways to prevent, detect, and extinguish them. In addition, advances into fundamental combustion		

	configurations are anticipated.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study is of relevance to fire safety in terrestrial environments where oxygen enhanced combustion is used and in emergency rooms in hospitals etc. where oxygen storage and use is common.
Task Progress:	The Principal Investigator team at IUPUI has been developing its combustion laboratory. An enclosure, an NDF burner and an IDF burner and flow system were built for the experiments by a graduate student at IUPUI. A digital camera was purchased and used to image NDFs and IDFs. The student did his summer internship at NASA Glenn under Dr. Sunderland's supervision. The internship provided a valuable training opportunity for the student. Development of a numerical code to predict the analytical Roper flame shape this year has resulted in establishment of empirical parameters based on microgravity and normal gravity flames. Reasonable agreement is now being obtained between the Roper predictions and observed shapes of ethane NDFs and IDFs in normal gravity and microgravity. A new PhD student has been hired at IUPUI to continue soot extinction and radiometric measurements on flames run by the previous student and some new test conditions for producing sooting flames. An undergraduate student has also been working on an independent project to design a thermophoretic sampling probe for soot for these flames. The Co-Investigator team at Purdue (West Lafayette) has performed global chemistry and detailed chemistry computations (involving 99 species and 1066 reaction mechanisms including PAH) for the NDF and IDF configurations for which experimental data on flame shapes are available. A Ph.D. student was hired to support this effort. Some local acceleration effects noticed for the IDFs in the global chemistry calculations were confirmed to be present in detailed chemistry calculations. Domain and grid independence studies were completed for global chemistry calculations for the test cases. The results of the investigations are being compared with the experimental and analytical results and being compiled into a journal paper for publication. The model will include multi-ray, coupled radiation heat transfer calculations.
Bibliography Type:	Description: (Last Updated: 03/10/2008)
Abstracts for Journals and Proceedings	Krishnan SS, Abshire JM, Gore JP, Sunderland PB. "Comparisons of Model Predictions with Measurements of Normal and Microgravity Laminar Diffusion Flame Shapes" Spring Technical Meeting of the Combustion Institute Central States Section, Austin, TX, March 2004. Proceedings, Spring Technical Meeting of the Combustion Institute Central States Section, Austin, TX, March 2004, 5 pp. , Mar-2004
Abstracts for Journals and Proceedings	Bhatia P, Katta VR, Sunderland PB, Krishnan SS, Gore JP. "CFD Based Global Chemistry Predictions for Normal and Inverse Laminar Ethane Jet Diffusion Flames Under Oxygen Enhancement and Gravity Variation." Spring Technical Meeting of the Combustion Institute Central States Section, Austin, TX, March 2004. Proceedings, Spring Technical Meeting of the Combustion Institute Central States Section, Austin, TX, March 2004, 6 pp. , Mar-2004
Articles in Peer-reviewed Journals	Sunderland PB, Krishnan SS, Gore JP. "Effects of oxygen enhancement and gravity on normal and inverse laminar jet diffusion flames." Combustion and Flame, 2004 Jan;136(1-2):254-6. http://dx.doi.org/10.1016/j.combustflame.2003.09.015 , Jan-2004
Awards	Abshire JM. "Graduate Presentation Award - Second Place at the Annual Region III Student Conference for his presentation of "Normal and Microgravity Laminar diffusion Flame Shapes--Anayltical Models and Comparisons," March 2004." Mar-2004
Presentation	Krishnan, S.S.; Abshire, J.M.; Yuan, Z.-G.; Sunderland, P.B.; Gore, J.P. "Normal and Microgravity Laminar Diffusion Flame Shapes – Analytical Models and Comparisons" AIAA Annual Region III Student Conference Mar-2004